ComNetX: Local Hierarchical Adaptation for Batched Dynamic Community Detection

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ABSTRACT Rerunning a community detector on a dynamic graph after every batch of edge updates can be expensive. ComNetX adapts existing detectors by selecting communities that intersect a neighborhood of the affected endpoints. It contracts vertex groups defined by a stored hierarchy, runs the detector on the resulting weighted graph, and maps the partition back to the original vertices. The first pass updates the reported finest partition. Subsequent passes update the coarser partitions while preserving nesting. On the selected induced subgraph, contraction preserves modularity for partitions that keep each contracted group intact. We analyze time and memory costs and evaluate update streams constructed from six real graphs. On five graphs initialized from approximately 99.9% of the stored records, warm-started Leiden on the full graph took 6.60 to 98.72 times as long as ComNetX with Leiden. Absolute differences in final modularity ranged from 0.0035 to 0.0213. Over 500 updates covering the final 10% of the records on PubMed and ArxivMath, the corresponding time ratios were 24.46 and 72.67, respectively, and the absolute modularity difference stayed below 0.008 at every update. Speedups were smaller when the graphs grew from singleton partitions. In 20 paired runs with a feature-aware detector, ComNetX also reduced processing time under the different cluster-count rules used for the full and local graphs.

INDEX TERMS community detection, dynamic graphs, graph clustering, hierarchical graph contraction, incremental graph analytics

I. INTRODUCTION†

COMMUNITY structure helps characterize communication networks [1] and the evolution of social groups [2]. As edges change, maintaining a partition requires deciding which assignments to revisit [3]. Even a small batch can trigger a costly full-graph computation with a modularity heuristic such as Louvain or Leiden [4], [5], or with an attributed method that constructs features or trains a model [6]–[8].

A static detector can start afresh on every snapshot or use the preceding partition, as in the unseeded and seeded Grappolo schemes [9]. Incremental methods save work by screening vertices or changing how the optimizer processes them [10]–[13]. These rules depend on the underlying objective or optimizer. ComNetX instead changes the graph supplied to an existing detector. Its update region contains whole communities, and its stored hierarchy determines which vertices the detector can reassign individually. We make three contributions.

1) We construct a weighted quotient of each update region and lift the detector's result to the original vertices. The same procedure supports topology-only detectors and feature-aware detectors with aggregated attributes.
2) We prove that the update preserves nesting, apply the standard modularity-preserving contraction identity to each induced subgraph, and derive time and memory bounds, including the graph-wide worst case.
3) On streams from six real graphs, we compare quality and processing time against full Leiden recomputation and native dynamic frontier Leiden (DF-Leiden). The evaluation includes 500-update sequences, graph growth, batch subdivision, and a feature-aware detector.

II. RELATED WORK†

A. COMMUNITY DEFINITIONS, OBJECTIVES, AND EVALUATION

Community definitions include criteria based on internal and external connectivity [14]–[17]. Modularity compares within-community edge weight with a degree-preserving null model [18], [19]. Normalized cut measures separation relative to group volume [20], the map equation uses random-walk

compression [21], and stochastic blockmodels specify edge probabilities between latent groups [22].

For modularity optimization, Louvain alternates local moves and aggregation [4]. Leiden adds refinement to address poorly connected communities [5]. Other static methods use shared-memory parallelism [23], [24], fast label propagation [25], modularity-guided label updates [26], greedy modularity disassembly [27], or iterated local moves [28].

Modularity has resolution effects and many near-optimal partitions [29], [30]. Algorithm rankings also depend on graph structure and evaluation design [31]. Because reference class labels can reflect a different organizing principle from structural communities [32], [33], we report modularity and normalized mutual information (NMI) with these labels separately [34].

B. INCREMENTAL AND PARALLEL COMMUNITY DETECTION

Incremental methods maintain partitions across snapshots [3] using batch-update rules [35], dynamic label propagation [36], [37], or modularity-based updates [10], [38]. Delta-Screening uses modularity tests to select vertices for reconsideration [11]. C-Blondel compresses previously detected communities into supernodes and applies Louvain to the compressed graph [39].

Dynamic-frontier methods expand the set of vertices reconsidered by the optimizer [12], [40]. Parallel approaches include the shared-memory DyComPar algorithm [41] and the distributed DyG-DPCD algorithm [42]. Maintaining graph properties under insertions and deletions is also the subject of fully dynamic graph algorithms [43].

For Leiden, Sahu et al. combine affected-vertex processing with selective refinement and changes to community renumbering and aggregation [13]. Subsequent work tracks community identifiers and uses accumulated changes in community edge weight to guide refinement [44]. LD-Leiden repairs a stored hierarchy around an affected frontier [45]. HIT-Leiden maintains connected components and hierarchical community structures to restrict the affected range [46]. These approaches modify the Leiden optimizer. ComNetX maintains its own hierarchy and can use any compatible detector to partition a weighted quotient.

C. TEMPORAL MODELS AND LOCAL OUTPUT REQUIREMENTS

NOME balances snapshot quality and temporal smoothness [47]. Dynamic stochastic blockmodels use a state-space model [48], and flow stability accounts for temporal edge ordering [49]. Other approaches model community evolution through the Matthew effect [50] or track communities through modularity [51]. Continuous-time L-modularity avoids choosing snapshot boundaries [52]. Further work addresses sparse temporal observations [53] and random-walk snapshot clustering [54]. ComNetX updates a snapshot partition without adding a temporal term to the detector's objective.

Seed-, query-, and anchor-based methods retrieve nearby communities [55]–[57]. OLCPM maintains overlapping communities online [58]. ComNetX maintains a complete, nonoverlapping partition and keeps community assignments outside the update region unchanged.

D. ATTRIBUTED AND REPRESENTATION-LEARNING METHODS

Attributed community detection combines topology with vertex features [59]. Graph neural networks learn representations of graph-structured inputs [60]. SEAL learns detection heuristics [61], CommDGI uses Deep Graph Infomax [62], and DMoN and DGCluster use neural modularity objectives [6], [63]. MAGI connects modularity maximization with contrastive learning [7]. Comparative studies and surveys examine these choices of representations, losses, and evaluation criteria [64], [65].

DAG learns communities without a prescribed cluster count [66]. Other methods use deep structural entropy [67] or incorporate group-sensitive connectivity in fairness-aware clustering [68]. Work on efficiency includes pre-training followed by refinement [69] and inductive graph partitioning [70]. We use Spectral Subspace Clustering for Attributed Graphs (S^2CAG) [8] to evaluate how the detector processes features aggregated over contracted vertices.

Kong et al. combine matrix factorization clustering (MFC) with TopoReg to encourage consistent inter-community structure across snapshots [71]. Community structure also supports dynamic popularity prediction [72]. Applying a detector to a quotient graph does not in itself preserve a learned representation or a temporal objective.

III. PROBLEM FORMULATION†

A. BATCHED GRAPH MODEL

Let $G_t = (V, E_t, \boldsymbol{A}_t, \boldsymbol{X})$ be the graph after batch t, with a fixed vertex set V, $n = |V|$, sparse adjacency $\boldsymbol{A}_t$, and optional node-feature matrix $\boldsymbol{X} \in \mathbb{R}^{n \times d}$. A sparse signed batch $\boldsymbol{\Delta}_t$ updates the stored adjacency as

$$\boldsymbol{A}_t = \boldsymbol{A}_{t-1} + \boldsymbol{\Delta}_t. \tag{1}$$

The affected endpoints are

$$S_t = \{i : \exists j,\ \Delta_t(i,j) \neq 0 \text{ or } \Delta_t(j,i) \neq 0\}. \tag{2}$$

We require $\boldsymbol{A}_t$ to remain nonnegative. The experiments use insertions and positive weight increments with fixed vertex attributes.

A compatible detector returns a partition through the interface

$$\mathcal{B}(\boldsymbol{A}, \boldsymbol{X}, \boldsymbol{y}_0; \boldsymbol{\theta}) \longmapsto \boldsymbol{y}, \tag{3}$$

where $\boldsymbol{X}$ and the initial partition $\boldsymbol{y}_0$ are optional, and $\boldsymbol{\theta}$ contains settings such as resolution or an iteration budget. The output has one integer label per input vertex. Vertices with the same label form a community. A native dynamic

implementation can serve as a backend if it can partition each supplied quotient independently.

Full-snapshot recomputation supplies the entire adjacency $\boldsymbol{A}_t$ and, when supported, the preceding partition:

$$\boldsymbol{y}_t^{\text{full}} = \mathcal{B}(\boldsymbol{A}_t, \boldsymbol{X}, \boldsymbol{y}_{t-1}^{\text{full}}; \boldsymbol{\theta}). \tag{4}$$

The full Leiden baseline uses this warm start. Section IV defines the graphs supplied to the detector by ComNetX.

B. PARTITION QUALITY

The paired comparisons use the same undirected representation of each graph, symmetrizing directed source graphs. For a weighted adjacency, let $W = \sum_{i,j} A_{ij} > 0$, $d_i^{\text{out}} = \sum_j A_{ij}$, and $d_j^{\text{in}} = \sum_i A_{ij}$. For a partition C, we compute directed modularity at resolution γ as

$$Q_\gamma(\boldsymbol{A}, C) = \frac{1}{W} \sum_{i,j} \left(A_{ij} - \gamma \frac{d_i^{\text{out}} d_j^{\text{in}}}{W} \right) \mathbf{1}\{C(i) = C(j)\}, \tag{5}$$

which reduces to the standard undirected expression when $\boldsymbol{A}$ is symmetric and $d_i^{\text{out}} = d_i^{\text{in}} = d_i$. The main experiments use $\gamma = 1$. We also use normalized mutual information (NMI),

$$\text{NMI}(C, Z) = \frac{2I(C; Z)}{H(C) + H(Z)}, \tag{6}$$

where I denotes mutual information, H denotes entropy, and Z contains the reference class labels [34].

IV. COMNETX METHOD†

A. UPDATE PROCEDURE AND STORED STATE

At level ℓ, ComNetX partitions the update region U_t^ℓ into indivisible groups, called *atoms*, that form a partition $\mathcal{G}_t^\ell$. The detector receives their quotient adjacency $\bar{\boldsymbol{A}}_\ell$. The update regions are fixed before the first pass and processed from fine to coarse (Fig. 1).

ComNetX stores the sparse adjacency $\boldsymbol{A}_t$, optional features $\boldsymbol{X}$, and an $L \times n$ array $\boldsymbol{C}_t = (C_t^0, \ldots, C_t^{L-1})$ of integer community labels. Entry $C_t^\ell(v)$ is the label of vertex v at level ℓ after batch t. Let $\mathcal{P}_t^\ell$ be the corresponding partition. We write $\mathcal{P} \preceq \mathcal{Q}$ if every block of $\mathcal{P}$ lies in a block of $\mathcal{Q}$. The finest partition, $\mathcal{P}_t^0$, is reported. The coarser partitions determine contraction groups for later batches. Each block H has the canonical label $\kappa(H) = \min H$.

In the maintenance experiments, Leiden initializes $\mathcal{P}_0^0$ on the full initial graph, including for feature-aware detectors. Each coarser partition is obtained by partitioning the quotient of the preceding level and lifting its labels to V. This gives a nested initial hierarchy in which adjacent levels may coincide. The growth control initializes all levels with singleton blocks.

B. SELECTING THE UPDATE REGIONS

After applying a batch, ComNetX expands S_t by r hops on the support of $\boldsymbol{A}_t + \boldsymbol{A}_t^\mathsf{T}$. Nonnegativity ensures that the sparse matrix–vector traversal computes reachability on this symmetrized support. Let

$$B_t = N_r(S_t) = \{v : \text{dist}(v, S_t) \le r\}. \tag{7}$$

Before any partition is modified, an update region is computed at each level from the previous hierarchy:

$$U_t^\ell = \bigcup_{\substack{H \in \mathcal{P}_{t-1}^\ell \\ H \cap B_t \neq \varnothing}} H, \qquad 0 \le \ell < L. \tag{8}$$

Each finer block meeting B_t belongs to a coarser block also meeting B_t. Hence

$$B_t \subseteq U_t^0 \subseteq U_t^1 \subseteq \cdots \subseteq U_t^{L-1}. \tag{9}$$

C. CONTRACTION AND PARTITION UPDATE

1) Atoms and Quotient Construction

The first pass, at the reported level 0, uses the base atoms

$$\begin{aligned} \mathcal{G}_t^0 = \{\{v\} : v \in B_t\} \\ \cup \{(H \cap U_t^0) \setminus B_t : H \in \mathcal{P}_{t-1}^{L-1}, \\ (H \cap U_t^0) \setminus B_t \neq \varnothing\}. \end{aligned} \tag{10}$$

Thus vertices in B_t can be reassigned individually. Each remaining atom may span several fine communities within one pre-update coarsest community. At every coarser pass, the atoms are the complete blocks of the already updated finer partition within the update region:

$$\mathcal{G}_t^\ell = \{H \in \mathcal{P}_t^{\ell-1} : H \subseteq U_t^\ell\}. \tag{11}$$

Every pass starts from a fresh restriction of the stored adjacency,

$$\boldsymbol{A}_\ell^{\text{work}} = \boldsymbol{D}_\ell \boldsymbol{A}_t \boldsymbol{D}_\ell, \tag{12}$$

where $\boldsymbol{D}_\ell = \text{diag}(\mathbf{1}_{U_t^\ell})$ zeros rows and columns outside U_t^ℓ. Restricting the indices to U_t^ℓ gives the induced adjacency $\boldsymbol{A}_t[U_t^\ell]$. Let $k_\ell = |\mathcal{G}_t^\ell|$. In the membership matrix $\boldsymbol{P}_\ell \in \{0,1\}^{k_\ell \times n}$, the column for each vertex in U_t^ℓ has a single unit entry in the row of its atom. Columns outside U_t^ℓ are zero. The quotient adjacency is

$$\bar{\boldsymbol{A}}_\ell = \boldsymbol{P}_\ell \boldsymbol{A}_\ell^{\text{work}} \boldsymbol{P}_\ell^\mathsf{T}. \tag{13}$$

The backend therefore receives one vertex per atom and the total weight between every ordered pair of atoms.

For a feature-aware backend, ComNetX supports sum and mean aggregation. For $G_{\ell a} \in \mathcal{G}_t^\ell$,

$$\bar{\boldsymbol{x}}_{\ell a} = \begin{cases} \displaystyle\sum_{i \in G_{\ell a}} \boldsymbol{x}_i, & \text{sum}, \\ \displaystyle\frac{1}{|G_{\ell a}|} \sum_{i \in G_{\ell a}} \boldsymbol{x}_i, & \text{mean}. \end{cases} \tag{14}$$

2) Lifting the Partition

The detector partitions the quotient without initial labels. Each atom's vertices inherit its returned label through $\boldsymbol{P}_\ell$. Each resulting block $H \subseteq U_t^\ell$ receives the canonical label $\min H$. Assignments outside U_t^ℓ remain unchanged (Algorithm 1).

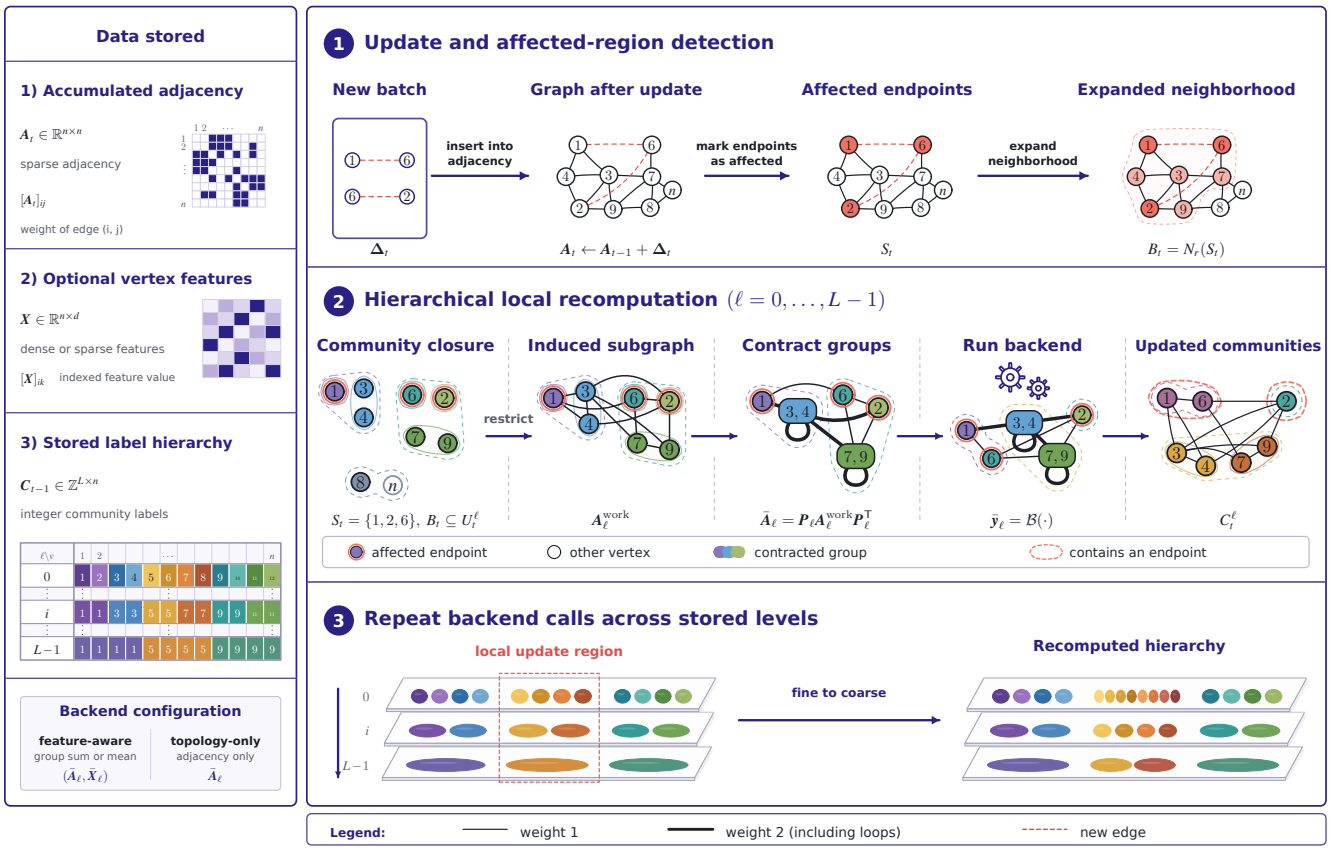


FIGURE 1. **Local recomputation after insertion of edges (1, 6) and (6, 2) (Algorithm 1). The middle panel shows the operations at each level $\ell = 0, \ldots, L-1$. The nested outlines in the closure diagram show the pre-update hierarchy. At level 0, (10) gives each vertex in B_t its own atom and groups the remaining vertices of U_t^0 by their coarsest pre-update community. At each coarser level, (11) contracts each complete block of the updated finer partition within U_t^ℓ to one quotient vertex. In the coarser pass illustrated here, the detector merges the quotient vertices for $\{3, 4\}$ and $\{7, 9\}$, giving all four original vertices the same label. Vertices 8 and n lie outside the update region and retain their labels. The stored arrays and the hierarchy in the lower panel are schematic.**

D. STRUCTURAL PROPERTIES

1) Hierarchy Consistency

Starting from a canonical, nested hierarchy, Algorithm 1 preserves nesting:

$$\mathcal{P}_t^0 \preceq \mathcal{P}_t^1 \preceq \cdots \preceq \mathcal{P}_t^{L-1}. \tag{15}$$

For $\ell > 0$, the pre-update region U_t^ℓ is a union of complete blocks at level $\ell - 1$ by nesting. The finer pass changes only $U_t^{\ell-1} \subseteq U_t^\ell$ by (9). Its newly formed blocks stay inside $U_t^{\ell-1}$, and their canonical labels are disjoint from those of the unchanged outside blocks. Thus U_t^ℓ remains a union of complete updated finer blocks. By (11), each is indivisible at level ℓ. The outside blocks retain their previous nesting. Induction over ℓ proves (15).

2) Modularity under Contraction

We apply the standard modularity-preserving contraction [73] to a region U. Let $\boldsymbol{A}_U$ be its induced adjacency with total weight $W_U > 0$. For a partition $\mathcal{G} = \{G_1, \ldots, G_k\}$ of U with membership matrix $\boldsymbol{P}_U \in \{0,1\}^{k \times |U|}$, the quotient $\bar{\boldsymbol{A}} = \boldsymbol{P}_U \boldsymbol{A}_U \boldsymbol{P}_U^\top$ retains all self-loops and has entries

$$\bar{A}_{ab} = \sum_{i \in G_a} \sum_{j \in G_b} A_{ij}. \tag{16}$$

Consequently, the quotient preserves total weight and the sums of outgoing and incoming weighted degrees. Every quotient partition Z induces a unique partition C of U that keeps each atom intact, and, under the same directed or symmetric adjacency, self-loop, and resolution conventions,

$$Q_\gamma(\bar{\boldsymbol{A}}, Z) = Q_\gamma(\boldsymbol{A}_U, C). \tag{17}$$

Expanding the internal-weight and degree terms in (5) gives the identity. It applies to modularity for partitions that keep the atoms intact. Other objectives, including those using features, require separate analysis.

The induced graph omits crossing edges and uses a local normalization, so its modularity ranking can differ from the full-graph ranking. Appendix A gives the exact difference. We evaluate the reported finest partition on the full graph.

Algorithm 1 ComNetX update for one batch

Require: $\boldsymbol{A}_{t-1}, \boldsymbol{C}_{t-1}$, optional $\boldsymbol{X}$, update $\boldsymbol{\Delta}_t$, radius r, backend $\mathcal{B}$, settings $\boldsymbol{\theta}$

$\boldsymbol{A}_t \leftarrow \boldsymbol{A}_{t-1} + \boldsymbol{\Delta}_t$
$S_t \leftarrow$ endpoints of the updated edges; $B_t \leftarrow N_r(S_t)$
for $\ell = 0, \dots, L-1$ **do**
  $U_t^\ell \leftarrow$ closure of B_t under $\mathcal{P}_{t-1}^\ell$ ▷ fixed before all passes
end for
$\boldsymbol{C}_t \leftarrow \boldsymbol{C}_{t-1}$
for $\ell = 0, \dots, L-1$ **do** ▷ fine to coarse
  if $\ell = 0$ **then**
    $\mathcal{G}_t^\ell \leftarrow$ singleton atoms for B_t and the nonempty residual groups in (10)
  else
    $\mathcal{G}_t^\ell \leftarrow$ complete updated $\mathcal{P}_t^{\ell-1}$-blocks in U_t^ℓ
  end if
  $\boldsymbol{A}_\ell^{\mathrm{work}} \leftarrow \boldsymbol{D}_\ell \boldsymbol{A}_t \boldsymbol{D}_\ell$
  build $\boldsymbol{P}_\ell$ from $\mathcal{G}_t^\ell$
  $\bar{\boldsymbol{A}}_\ell \leftarrow \boldsymbol{P}_\ell \boldsymbol{A}_\ell^{\mathrm{work}} \boldsymbol{P}_\ell^{\mathsf{T}}$
  aggregate $\bar{\boldsymbol{X}}_\ell$ if $\mathcal{B}$ uses features
  $\bar{\boldsymbol{y}}_\ell \leftarrow \mathcal{B}(\bar{\boldsymbol{A}}_\ell, \bar{\boldsymbol{X}}_\ell, \varnothing; \boldsymbol{\theta})$
  lift $\bar{\boldsymbol{y}}_\ell$ to U_t^ℓ; set $C_t^\ell(v) \leftarrow \min H$ for every lifted block H and $v \in H$
end for
return $\boldsymbol{C}_t$ with reported partition C_t^0

E. TIME AND MEMORY COMPLEXITY

For $r = 1$, let $d_t(v)$ denote the degree of v in the symmetrized support graph. Then

$$|B_t| \le |S_t| + \sum_{v \in S_t} d_t(v). \tag{18}$$

Thus, a small edge batch can still have a large radius-one expansion when it touches hubs. If $\mathcal{P}_{t-1}^\ell$ is the pre-update partition at level ℓ, the update-region size $h_\ell = |U_t^\ell|$ is exactly

$$h_\ell = \sum_{H \in \mathcal{P}_{t-1}^\ell : H \cap B_t \neq \varnothing} |H|. \tag{19}$$

Let $r_t \le r$ be the number of neighborhood traversals, s_t the number of stored adjacency entries, e_ℓ the number of entries in $\boldsymbol{A}_t[U_t^\ell]$, and k_ℓ and q_ℓ the numbers of quotient vertices and stored entries in the quotient adjacency matrix. Let $\overline{T}_{\mathcal{B}}(k,q,d)$ and $\overline{M}_{\mathcal{B}}(k,q,d)$ be nondecreasing worst-case envelopes for the core backend time and working memory. We account for format conversion separately, assuming its cost is linear in the stored adjacency and feature arrays. Appendix B treats the current implementation.

1) Worst-Case Resources

Let $F = nd$ bound dense or sparse feature storage for feature-aware calls and $F = 0$ for topology-only calls. Suppose levels are processed sequentially, with quotient edges constructed by direct endpoint relabeling followed by comparison-based sort-and-reduce. After applying the batch, an update requires at most

$$T_{\mathrm{update}} = O\Big(r_t(n+s_t) + L\big[(n+s_t)\log(2+n+s_t) + F\big] + \sum_{\ell=0}^{L-1} \overline{T}_{\mathcal{B}}(k_\ell, q_\ell, d)\Big) \tag{20}$$

time and

$$M_{\mathrm{peak}} = O\Big(s_t + Ln + F + \max_\ell\{e_\ell + h_\ell + q_\ell + \overline{M}_{\mathcal{B}}(k_\ell, q_\ell, d)\}\Big) \tag{21}$$

memory. The time bound includes r_t sparse traversals and, at each level, scans of the label and edge arrays, sorting at most s_t entries, feature aggregation, format conversion, and a backend call. Beyond the $O(s_t + Ln + F)$ stored state, only one level and its backend workspace are needed at a time.

Since $h_\ell, k_\ell \le n$ and $e_\ell, q_\ell \le s_t$, a nondecreasing backend cost gives the explicit graph-wide worst case

$$T_{\mathrm{worst}} = O\Big(r_t(n+s_t) + L\big[(n+s_t)\log(2+n+s_t) + F + \overline{T}_{\mathcal{B}}(n, s_t, d)\big]\Big). \tag{22}$$

An update touching a hub or a large community can make each of the L quotient graphs comparable in size to the full snapshot.

2) Costs of the Present Implementation

The PyTorch implementation uses sparse tensor operations in place of the sort-and-reduce construction assumed in (20). It scans the full label and edge arrays at each level. When it executes r_t traversals and all L passes, it incurs

$$\Omega(r_t s_t + L(n+s_t)) \tag{23}$$

full-scan memory traffic even for small quotients. Appendix B gives the temporary-memory requirements. Local recomputation saves time when the reduction in detector work exceeds the cost of selecting, contracting, and lifting the update region.

V. EXPERIMENTAL DESIGN†

We organize the evaluation around five questions.

RQ1 What quality–time trade-off does ComNetX achieve relative to warm-started full-snapshot recomputation?

RQ2 How does this trade-off change with the wrapped backend, including feature-aware calls under backend-specific interface rules and a specialized dynamic implementation?

RQ3 How do radius expansion and community closure change the update region, and how does quotient contraction determine the final backend input?

TABLE 1. Source graphs and actual updates under 999:10. Source counts are undirected edges for Cora, ACM, and CiteSeer, and directed arcs for the other graphs.

Dataset	Vertices	Source count	Order	Updates
Cora	2,708	5,278	artificial	10
ACM	3,025	13,128	artificial	10
CiteSeer	3,327	4,552	artificial	8
Patent	12,214	41,916	timestamp	10
PubMed	19,717	44,338	artificial	10
ArxivMath	270,013	799,745	timestamp	10

RQ4 How sensitive is the measured quality–time trade-off to hierarchy depth, radius, feature representation, aggregation, and random-feature initialization?

RQ5 How does the trade-off evolve over longer update sequences, and how does it depend on update granularity and the fraction of the graph already available at initialization?

A. DATASETS AND STREAM CONSTRUCTION

We use six graphs (Table 1). Patent and ArxivMath supply timestamped records, which we process in chronological order [74], [75]. Cora, ACM, CiteSeer, and PubMed are static attributed graphs, so we use a fixed artificial order for each stream. The first three are available from the collection described in [65], and the Cora and CiteSeer topologies match the preprocessed Planetoid graphs [76]. For PubMed we use the 44,338-arc version distributed by PyTorch Geometric.[1]

We form batches from stored records. Cora, ACM, and CiteSeer store both orientations of each edge. When they arrive in different batches, the first record inserts the undirected edge and the second increases its weight. Full and local runs use the same weighted snapshots.

In protocol p:q, we divide the stream into $p+1$ roughly equal parts, use the first p to build the initial graph, and process the last part in up to q batches. Thus 999:10 tests maintenance after approximately 99.9% of the records have arrived. Varying q changes the batch size over the same tail. The 99:50 and 9:500 protocols reserve larger tails (about 1% and 10%, respectively), while 0:10 measures growth from singleton partitions over the entire stream.

Table 2 summarizes the comparisons. The main S^{2}CAG study covers all six graphs. For Leiden we report the graphs with complete pairs of runs: the short maintenance series lacks a Patent run, and the growth series lacks a ComNetX result on ArxivMath.

B. TIMING AND COMPUTING ENVIRONMENT

We report cumulative processing time over the update stream (Table 3). For full recomputation and ComNetX, we measure wall-clock time and subtract the adapter's conversion and preprocessing interval. For native DF-Leiden, the binding reports time inside `apply()` after the exposed `update()` call. The time ratio is baseline time divided by ComNetX time. Ratios involving native DF-Leiden therefore compare two differently timed execution paths.

The original experiments ran on a Linux server with two 64-core AMD EPYC 7742 central processing units (CPUs), 2.0 TiB of memory, and eight NVIDIA A100 graphics processing units (GPUs) with 80 GB each. The repeated S^{2}CAG experiments used the same processor model and one visible A100. Graph tensors resided on the GPU and the spectral solver used eight processor threads. The two software environments are documented in the repository.

For the graph-size measurements (RQ3), we ran `leidenalg` on a CPU with $L = 3$, $r = 1$, and protocol 999:10. These runs supply the neighborhood, update-region, and quotient sizes for Cora, PubMed, and ArxivMath. All timing comparisons use the server runs.

C. METHODS AND FIXED CONFIGURATION

Leiden uses resolution one. Full recomputation starts from its previous partition. ComNetX uses $L = 3$ and $r = 1$ by default. It initializes the hierarchy with Leiden for maintenance and with singleton partitions for growth. Quotient calls receive no initial partition. Native DF-Leiden maintains its state across batches, whereas ComNetX starts a fresh detector on each quotient.[2]

LD-Leiden is the closest Leiden-specific alternative. Its initialization and timing interfaces differ from those of the paired benchmark, so we compare it qualitatively in Section II.

1) S^{2}CAG Solver Settings

The S^{2}CAG adapter uses the `sub` solver with propagation order $T = 10$, $\alpha = 1$, $\tau = 7$ singular value decomposition (SVD) iterations, ten oversampling columns, and 100 rounding iterations. The `sub` solver ignores `gamma`; we evaluate modularity externally at resolution one. Each invocation makes five solver calls and returns the last result. The SVD seed defaults to zero, and the internal standard deviation is computed from a single observation after the loop.

2) Requested and Realized Cluster Counts

Full S^{2}CAG uses the previous partition only to set the requested number of clusters k. Its spectral solver and rounding procedure receive no initial labels. A local call requests one cluster per quotient vertex. For an input with n vertices and d features, the slice `U[:, 1:k+1]` discards the leading singular vector and retains $s \le \min\{k, \min(n,d)-1\}$ columns. Rounding can leave columns empty, so a call may return fewer than s communities. The reported ComNetX partition also includes the communities outside the update region. The repeated experiments record the resulting group counts and labels.

[1] https://pytorch-geometric.readthedocs.io/en/latest/generated/torch_geometric.datasets.AttributedGraphDataset.html

[2] Original DF-Leiden implementation: https://github.com/puzzlef/leiden-communities-openmp-dynamic

TABLE 2. Experiments under 999:*q*, 99:50, 9:500, and 0:10. Each original timing comparison uses one execution per method and setting within its series. The repeated S²CAG study uses five fixed-input pairs per setting.

Purpose	Graphs	Protocol	Actual updates	Runs and controls
Main Leiden comparison	five real topologies	999:10	8–10	one paired execution
Feature-aware backend	six real topologies	999:10	8–10	dataset/random features
Repeated S²CAG pairs	PubMed, ArxivMath	999:5, 999:10	5, 10	five fixed-input repeats
Topology parameters	Cora, PubMed, ArxivMath	999:10	10	12 settings per graph
Graph-size diagnostics	Cora, PubMed, ArxivMath	999:10	10	instrumented runs
S²CAG feature controls	Cora, PubMed	999:10	10	modes, aggregation, radius
Feature-input sensitivity	six real topologies	999:10	8–10	feature seeds 1–5
Batch granularity	PubMed, ArxivMath	999:*q*	5–320	seven requested chunk counts
Larger update windows	PubMed, ArxivMath	99:50, 9:500	50, 500	one edge order per graph
Graph growth	five paired real topologies	0:10	10	singleton initialization

TABLE 3. Timing boundaries for all *p*:*q* protocols. "Excluded" means outside the timer or explicitly subtracted. In the graph-growth experiments there is no separate bootstrap; processing the first snapshot is included.

Stage	Full	ComNetX	Native DF
Data loading and initial feature generation	Excluded	Excluded	Excluded
Initial partition/bootstrap	Excluded	Excluded	Excluded
Applying the batch / exposed update call	Excluded	Excluded	Excluded
Neighborhood expansion and community closure	–	Included	Internal[a]
Induced subgraph, quotient, aggregation, lifting	–	Included	Internal[a]
Adapter conversion/preprocessing interval	Subtracted	Subtracted	–
Detector optimization	Included	All passes	Reported time[a]
S²CAG densification and five solver calls	Included	Included	–
S²CAG internal accuracy, NMI, and adjusted Rand index	Included	Included	–
External modularity and reference-label metrics	Excluded	Excluded	Excluded

[a]Native DF-Leiden reports the duration of `apply()`; the binding does not expose its internal stages.

3) Feature Inputs and Preprocessing

Both methods in each feature comparison use the same undirected graph. Unless specified otherwise, ComNetX uses mean aggregation. The original presets specify 64 random features per vertex with default seed 42, and the seed study varies that seed from 1 to 5. The loader draws independent standard normal entries once per graph using a CPU `torch.Generator`. The repeated campaign records the seed and hashes of the actual inputs.

Local S²CAG calls first aggregate signed features over atoms, then apply elementwise absolute value, followed by term frequency–inverse document frequency (TF–IDF) weighting and row L2 normalization. Positive and negative values can therefore cancel inside an atom before the absolute value is taken. Full calls apply the absolute value directly to the original features. The one-hot control supplies an identity feature matrix, whose rows are aggregated using the same atom membership.

4) Repeated Paired Experiments

A separate campaign evaluates PubMed and ArxivMath under 999:5 and 999:10, using five full/ComNetX pairs per setting. It uses the same fixed S²CAG solver settings, 64-dimensional random features with seed 42, and ComNetX with $L = 3$, $r = 1$, and mean aggregation. A shared initial hierarchy is constructed with public `leidenalg`, resolution one, seed one, and two iterations. The same initial hierarchy and record sequence are reused across the two subdivisions and all repeats. The initial finest partitions contain 55 communities on PubMed and 8,167 on ArxivMath.

Within each pair the methods run in separate processes, with their order alternating across pairs. Timing starts after warm-up and excludes batch application, external evaluation, and writing labels. We synchronize the device at the interval boundaries and report time both with and without adapter preprocessing. Both intervals include densification, the five solver calls, and their internal metrics. We report mean time and sample standard deviation across the five executions, and the median of the five paired time ratios.

VI. RESULTS AND DISCUSSION[†]

A. QUALITY AND PROCESSING TIME ON SHORT MAINTENANCE STREAMS (RQ1)

Under 999:10, warm-started full Leiden took 6.60 to 98.72 times as long as ComNetX (Table 4). ArxivMath had the largest time ratio: processing time fell from 110.4 to 1.118 s, and final modularity from 0.8858 to 0.8812.

Across the five graphs, absolute final modularity differences ranged from 0.0035 on CiteSeer to 0.0213 on ACM. Agreement with reference labels decreased on all five, with a larger drop on ArxivMath than on CiteSeer.

TABLE 4. Final quality and cumulative measured processing time for full-snapshot and ComNetX Leiden on 999:10 streams.

Dataset	Full Q	Local Q	Full NMI	Local NMI	Full T (s)	Local T (s)	Speedup
Cora	0.8240	0.8141	0.4667	0.4507	0.982	0.149	6.60
ACM	0.8255	0.8042	0.2700	0.2659	0.930	0.123	7.54
CiteSeer	0.8932	0.8897	0.3305	0.3304	0.647	0.077	8.43
PubMed	0.7872	0.7822	0.2067	0.1983	4.727	0.157	30.12
ArxivMath	0.8858	0.8812	0.4226	0.4070	110.4	1.118	98.72

B. RESULTS WITH DIFFERENT COMMUNITY-DETECTION BACKENDS (RQ2)

The native DF-Leiden comparison is reported with the sustained-maintenance experiments in Table 12. To evaluate the feature-aware interface, Table 5 reports S^{2}CAG on all six graphs with dataset and random features. With dataset features, ComNetX reduces measured processing time on five graphs, including a 51.72-fold reduction on ArxivMath from 4151.6 s to 80.270 s. Final NMI increases on five graphs as well.

The full and local S^{2}CAG calls use the cluster-count rules specified in Section V. Their time ratios consequently describe the implemented interface rather than equal-cluster-count optimization. With random features, full-run modularity is near zero, whereas ComNetX retains substantially higher values. The local procedure combines the feature input with graph structure and the stored hierarchy; the comparison does not attribute the quality difference to features alone. Patent with dataset features is the exception to the time reduction: ComNetX takes 13.474 s versus 11.390 s, while increasing both modularity and NMI.

1) Repeated S^{2}CAG Comparisons

Table 6 reports the separate repeated campaign. Every pair is faster with ComNetX under both timing definitions. With preprocessing subtracted, median speedups are 3.44 and 3.37 on PubMed and 37.63 and 70.40 on ArxivMath for 999:5 and 999:10, respectively. Including preprocessing gives 3.52, 3.46, 37.64, and 70.34 in the same order. Thus the time advantage in this campaign is not an artifact of subtracting preprocessing. Table 5 reports a separate execution series.

On ArxivMath, the first full-graph update requests $k = 8167$ and returns 16 and 15 groups under 999:5 and 999:10, respectively. It accounts for 93.64% and 90.85% of cumulative full processing time. The implemented randomized SVD uses $k + 10$ projection columns before truncating the retained embedding to at most 63 columns. The aggregate speedups therefore depend strongly on the adapter's initial cluster-count request. As a post hoc timing diagnostic, summing only updates $2, \ldots, q$ for both methods gives median paired speedups of 3.06 and 7.46, respectively (3.12 and 7.65 with preprocessing included).

ComNetX achieved higher final modularity and NMI in all four settings. Full S^{2}CAG returned only 11 or 13 communities, compared with 206–392 for ComNetX on PubMed and more than 8,100 on ArxivMath. The two methods therefore operate at markedly different partition granularities.

For PubMed, increasing q from 5 to 10 raises mean cumulative ComNetX time from 3.526 to 6.020 s. For ArxivMath, ComNetX time instead falls from 101.379 to 56.371 s, while final Q decreases from 0.5724 to 0.5633 and NMI from 0.1909 to 0.1774. The ArxivMath time reduction is concentrated in solver calls at level zero: their mean cumulative time falls from 97.472 to 51.222 s, with quotient sizes of 4,327–4,976 and 2,128–3,049 vertices, respectively. The traces locate the reduction in solver processing of smaller quotients; they do not separate costs within the spectral solver.

C. SIZE OF THE CONTRACTED GRAPHS (RQ3)

Table 7 reports neighborhood, update-region, and quotient sizes from the separate CPU executions. On PubMed and ArxivMath, the update region at level 0 contains, on average, 64.3% and 78.7% of all vertices, although the radius-one neighborhoods contain only 154.9 and 782.0 vertices. Community closure can therefore select a substantial part of a graph even for small batches. Contraction reduces these regions to an average of 167.9 and 825.3 vertices, respectively; the later passes use still smaller graphs. Thus a small contracted graph does not require a small update region.

The equality of k_1 and k_2 is present in every one of the 30 update records, before averaging. The corresponding update-region and quotient-edge counts also agree. These executions have coincident initial hierarchy levels, and the second pass does not further reduce the number of atoms supplied to the third pass. Equal adjacent levels are allowed; $k_1 = k_2$ is not a general property of ComNetX.

The quotient sizes describe the detector input. Selecting the region still requires scans of the full label and edge arrays (Section IV), so quotient size alone does not determine update time.

D. SENSITIVITY TO HIERARCHY DEPTH, RADIUS, AND FEATURES (RQ4)

Figure 2 shows all 36 tested depth–radius configurations. Every local point is faster than its full-snapshot baseline. The ranges are 8.52–23.42× on Cora, 7.63–125.54× on PubMed, and 15.27–636.84× on ArxivMath. The nondominated configurations differ across graphs in both hierarchy depth and radius.

TABLE 5. Main S²CAG feature comparison under 999:10; quality and time values are full-snapshot/ComNetX. Local calls use $L = 3, r = 1$ and mean aggregation. Each setting has one paired execution. Table 6 reports the separate repeated campaign.

Dataset	Features	Q	NMI	T (s)	Time ratio
Cora	dataset	0.6951/0.7859	0.4355/0.4504	29.949/5.553	5.39
ACM	dataset	0.0536/0.6219	0.0814/0.2538	224.2/10.713	20.93
CiteSeer	dataset	0.7770/0.7719	0.2542/0.3262	135.8/1.792	75.83
Patent	dataset	0.3970/0.5520	0.0604/0.3560	11.390/13.474	0.85
PubMed	dataset	0.6221/0.4987	0.1686/0.1524	78.869/13.609	5.80
ArxivMath	dataset	0.0003/0.0844	0.0002/0.1410	4151.6/80.270	51.72
Cora	random	-0.0020/0.7877	0.0233/0.4505	9.408/4.357	2.16
ACM	random	0.0004/0.6222	0.0089/0.2567	11.286/5.256	2.15
CiteSeer	random	-0.0031/0.7719	0.0170/0.3262	9.519/0.073	129.81
Patent	random	-0.0055/0.6579	0.0009/0.3560	10.165/8.802	1.15
PubMed	random	0.0027/0.5010	0.0003/0.1580	18.272/6.199	2.95
ArxivMath	random	0.0012/0.5875	0.0002/0.1769	3889.9/63.597	61.16

TABLE 6. Repeated paired S²CAG experiments under 999:5 and 999:10, with random features, $L = 3$, $r = 1$, and mean aggregation. Q, NMI, and K are full/ComNetX final values; K counts nonempty communities in the reported partition of the entire graph. Each method returns identical Q, NMI, and K in its five repeats at a given setting. Times exclude adapter preprocessing and are mean ± sample standard deviation in seconds. S is the median of the five paired time ratios, not the ratio of the displayed mean times.

Dataset	Protocol	Q	NMI	K	Full T (s)	ComNetX T (s)	S
PubMed	999:5	0.0001/0.4693	0.0003/0.0869	11/206	11.995±0.101	3.526±0.063	3.44
PubMed	999:10	0.0001/0.4683	0.0003/0.1112	11/392	20.342±0.095	6.020±0.065	3.37
ArxivMath	999:5	0.0020/0.5724	0.0003/0.1909	13/8,119	3819.065±21.885	101.379±0.563	37.63
ArxivMath	999:10	0.0024/0.5633	0.0003/0.1774	13/8,116	3951.633±37.385	56.371±0.196	70.40

TABLE 7. Measured graph dimensions, averaged over ten updates in the separate CPU diagnostics (999:10, $L = 3$, $r = 1$). Parentheses give percentages of all n vertices. Here $h_0 = |U_t^0|$, k_ℓ is the number of contracted vertices at level ℓ, and q_0 counts stored adjacency entries, including self-loops and both orientations of symmetric edges. The final column combines k_1 and k_2, which coincide in all 30 update records of this series.

Dataset	$\lvert B_t\rvert$	h_0	k_0	q_0	$k_1 = k_2$
Cora	9.4	314.2 (11.6%)	11.6 (0.43%)	45.6	3.0
PubMed	154.9	12,686.8 (64.3%)	167.9 (0.85%)	1,393.9	13.0
ArxivMath	782.0	212,558.8 (78.7%)	825.3 (0.31%)	7,408.8	47.4

On PubMed, $L = 2, r = 0$ yields modularity 0.7847 at 72.88× speedup, compared with full modularity of 0.7872. The highest local modularity on this graph, 0.7851, occurs at $L = 2, r = 2$. On ArxivMath, the fastest configuration, $L = 1, r = 0$, reaches 636.84× with modularity 0.8792; the higher-quality $L = 2, r = 2$ configuration reaches modularity 0.8819. Cora attains its highest local modularity at $L = 3, r = 2$. Neither a larger radius nor a deeper hierarchy consistently improves modularity. The common $L = 3, r = 1$ setting is used for consistency across datasets; it is not selected as the best configuration for each graph.

1) Effect of Feature Representation

Table 8 reports a separate feature-ablation execution series with the same solver settings as Table 5. The overlapping dataset/random-feature settings on Cora and PubMed have identical final Q and NMI at stored precision, but independently recorded times. Each table uses the full and local runs from its own series; these timings are not pooled.

On Cora, ComNetX maintains similar modularity across all three feature modes, 0.7859–0.7877. Random features give near-zero modularity in the full calls, yet the corresponding ComNetX results remain close to the local dataset-feature results: 0.7877 versus 0.7859 on Cora and 0.5010 versus 0.4987 on PubMed. With the one-hot input mode, ComNetX reduces the PubMed processing time from 1973.0 s to 120.0 s. These results show that the computational benefit can persist across the tested input representations, while the effect on partition quality depends on the graph and the backend's feature and cluster-count rules.

2) Feature Aggregation and Radius

The matching quality scores in Table 9 are expected under the S²CAG preprocessing used here. For each nonempty atom a, $\boldsymbol{x}_{\mathrm{mean}}(a) = \boldsymbol{x}_{\mathrm{sum}}(a)/|a|$. Taking absolute values preserves this positive scale factor and leaves the feature supports,

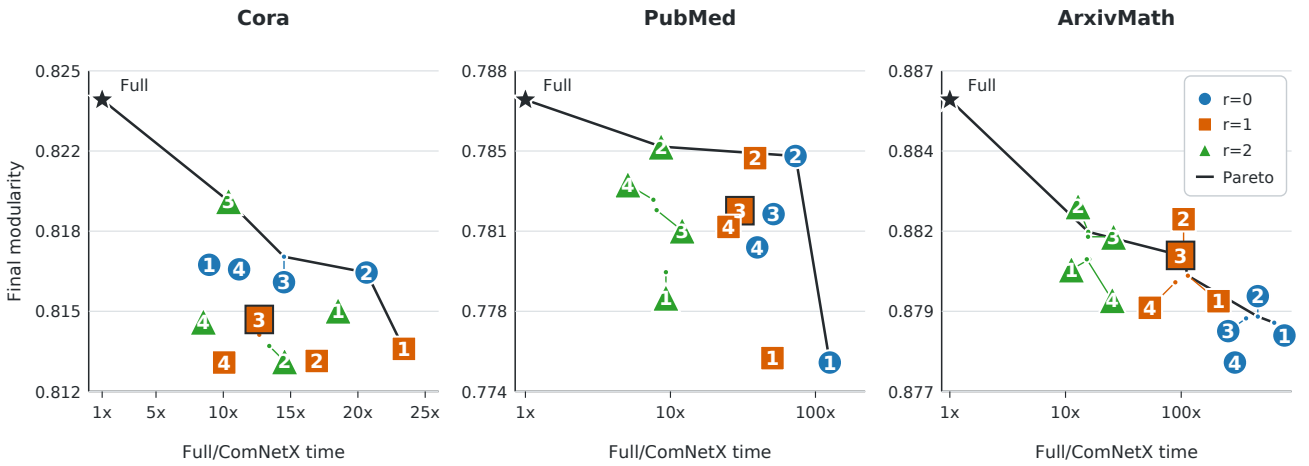


FIGURE 2. Effects of hierarchy depth and radius on Leiden modularity and processing time under 999:10, with $L \in \{1, 2, 3, 4\}$ and radius $r \in \{0, 1, 2\}$. Marker numbers show L; color and shape show r; and the black line connects nondominated measured points. The outlined orange square is the fixed $L = 3, r = 1$ configuration. Each point is one execution; ratios use the full baseline from the same parameter study.

TABLE 8. S²CAG feature sensitivity under 999:10, with ten updates on each graph. Each cell reports Q / NMI / T (seconds). These executions are separate from the main feature comparison in Table 5.

Variant	Cora	PubMed
Full, dataset features	0.6951/0.4355/30.156	0.6221/0.1686/78.119
ComNetX, dataset features	0.7859/0.4504/5.407	0.4987/0.1524/14.394
Full, random features	-0.0020/0.0233/8.571	0.0027/0.0003/18.861
ComNetX, random features	0.7877/0.4505/4.240	0.5010/0.1580/6.017
Full, one-hot input	0.6974/0.3528/35.310	0.6540/0.1071/1973.0
ComNetX, one-hot input	0.7862/0.4500/5.632	0.5349/0.1884/120.0

TABLE 9. S²CAG feature aggregation under 999:10, at $L = 3, r = 1$, using random features.

Dataset	Aggregation	Q	NMI	T (s)
Cora	mean	0.7877	0.4505	5.929
Cora	sum	0.7877	0.4505	4.197
PubMed	mean	0.5010	0.1580	6.177
PubMed	sum	0.5010	0.1580	5.995

TABLE 10. S²CAG radius sensitivity under 999:10, at $L = 3$, with random features and mean aggregation.

Dataset	r	Q	NMI	T (s)
Cora	0	0.8156	0.4509	1.740
Cora	1	0.7877	0.4505	4.368
Cora	2	0.6541	0.4120	4.829
PubMed	0	0.6601	0.1350	4.386
PubMed	1	0.5010	0.1580	6.248
PubMed	2	0.1659	0.0329	30.304

hence the IDF weights, unchanged. Linear TF–IDF weighting preserves the factor, and row L2 normalization cancels it; zero rows remain zero. Thus mean and sum yield the same normalized features in exact arithmetic, although floating-point rounding can introduce small differences. Each timing is based on one execution.

In Table 10, radius zero yields the highest modularity on both graphs. Increasing the radius to two reduces PubMed modularity from 0.6601 to 0.1659. Radius changes both the selected vertices and their aggregated representation; its effect on quality must therefore be assessed alongside its computational cost.

3) Variation across Random-Feature Seeds

Across the five random-feature seeds (Table 11), the sample standard deviation of final modularity is below 0.002 on Cora, ACM, CiteSeer, and Patent. It is 0.0379 on PubMed and 0.0665 on ArxivMath. PubMed shows the largest variation in NMI.

E. LONGER STREAMS, UPDATE GRANULARITY, AND INITIALIZATION (RQ5)

1) Sustained Maintenance over 500 Updates

ComNetX retains a substantial time advantage when a tenth of the stored record sequence is processed over 500 updates. Under 9:500, Leiden processing time falls from 248.6 s to 10.165 s on PubMed and from 5652.7 s to 77.788 s on Arxiv-

TABLE 11. ComNetX with S²CAG and random features under 999:10: mean ± sample standard deviation across feature seeds 1–5 on each fixed graph stream, with $L = 3, r = 1$ and mean aggregation.

Dataset	Q	NMI	T (s)
Cora	0.7853±0.0014	0.4499±0.0012	5.971±0.038
ACM	0.6221±0.0018	0.2571±0.0010	5.229±0.046
CiteSeer	0.7719±0.0000	0.3262±0.0000	1.753±0.004
Patent	0.6576±0.0004	0.3563±0.0002	8.676±0.101
PubMed	0.4481±0.0379	0.1241±0.0197	6.146±0.083
ArxivMath	0.5156±0.0665	0.1671±0.0087	62.490±1.990

Math, corresponding to 24.46-fold and 72.67-fold speedups. Table 12 reports final quality and time for both topology backends. For Leiden, the final modularity gaps are 0.0078 on PubMed and 0.0062 on ArxivMath. PubMed NMI increases from 0.2017 to 0.2045; ArxivMath NMI changes from 0.4205 to 0.4006. Thus PubMed's higher NMI accompanies a small modularity decrease, whereas both scores decrease on ArxivMath.

Fig. 3 shows that the absolute modularity difference remains below 0.008 at every measured update on both graphs. The mean local-minus-full difference is -0.00584 on PubMed and -0.00524 on ArxivMath; the minima are -0.00790 at update 477 and -0.00617 at update 500, respectively. The cumulative time ratio remains above one throughout both trajectories.

Against native DF-Leiden, the time advantage depends on the graph. ComNetX with DF-Leiden reduces ArxivMath time from 134.4 s to 13.517 s, while increasing final modularity from 0.8551 to 0.8792 and NMI from 0.3884 to 0.3967. On PubMed, a 0.354 s increase in processing time accompanies higher modularity (0.7786 versus 0.7619) and NMI (0.2051 versus 0.1946).

2) Subdivision of a Fixed Update Window

Table 13 varies the number of chunks in the final 0.1% of each stored record sequence. Finer subdivision increases the number of full-graph recomputations while reducing the number of records per update. The $q = 10$ row repeats the nominal short-maintenance setting in a separate execution series and uses its own full baseline. Thus PubMed's time ratio of 31.42 here differs from 30.12 in Table 4, although final Q and NMI coincide. This difference does not represent a change in parameters. On ArxivMath, the Leiden speedup grows from 64.12 to 695.84 as the requested count increases from 5 to 320; the final modularity gap increases from 0.0039 to 0.0134. The result identifies a favorable regime for local maintenance, but the larger ratio also reflects more baseline calls over the same set of stored records. It is not an increase in the amount of graph evolution tested.

PubMed has only 44 tail records, so requests for 80, 160, and 320 batches produce the same 44 updates. Final quality is identical, and the measured Leiden time ratios range from 45.43 to 45.57. Across all granularity settings, ComNetX with DF-Leiden has higher final modularity and NMI than the native stream on both graphs; its time ratio is 8.45–10.73 on ArxivMath and below one on PubMed.

3) Increasing the Updated Fraction

The 99:50 control processes approximately 1% of the stored records, between the 0.1% maintenance window and the 10% window of 9:500. Table 14 shows Leiden speedups of 24.16 on PubMed and 60.26 on ArxivMath, with final modularity differences of 0.0074 and 0.0041. DF-Leiden exhibits the same qualitative pattern as at 500 updates: the native stream is slightly faster on PubMed, while ComNetX is faster on ArxivMath and improves both quality measures on both graphs. Since both the initial snapshot and the number of calls change between protocols, these comparisons do not isolate update-window size as a single causal factor.

4) Growth from Singleton Initialization

In the growth control, both methods process the entire stream from singleton partitions. Table 15 reports the five complete paired graphs. ComNetX with Leiden is faster on all five, with ratios of 1.17–1.79 and final modularity differences of 0.0003–0.0080. NMI increases on Patent and PubMed. The time advantage is smaller than in the maintenance comparisons, where the graph and its initial hierarchy are already available.

VII. CODE AVAILABILITY AND LIMITATIONS†

The ComNetX implementation, experiment configurations, and archived results are available online.[3] The repository provides dataset preparation instructions. Rerunning S²CAG and native DF-Leiden requires external components not included in the artifact. The exact version of the measured DF-Leiden Python binding was not retained, limiting reproducibility of that baseline.

Four of the six graphs use artificial edge orders, and the window controls reuse the same record sequences. Most configurations have one execution per method. The repeated S²CAG study uses five runs per method and setting to assess runtime repeatability at fixed inputs. Independent temporal windows, edge deletions, arriving vertices, and changing node attributes have not been evaluated.

[3]https://github.com/mpailab/comnetx/tree/ieee-access-r14-20261108/artifacts/ieee-access

TABLE 12. Quality and measured processing time after 500 updates under 9:500. Values are baseline/ComNetX; the baseline is full Leiden or native dynamic DF-Leiden. The time ratio is baseline time divided by ComNetX time.

Dataset	Backend	Q	NMI	T (s)	Time ratio
PubMed	Leiden	0.7864/0.7786	0.2017/0.2045	248.6/10.165	24.46
PubMed	DF-Leiden	0.7619/0.7786	0.1946/0.2051	4.442/4.796	0.93
ArxivMath	Leiden	0.8860/0.8799	0.4205/0.4006	5652.7/77.788	72.67
ArxivMath	DF-Leiden	0.8551/0.8792	0.3884/0.3967	134.4/13.517	9.94

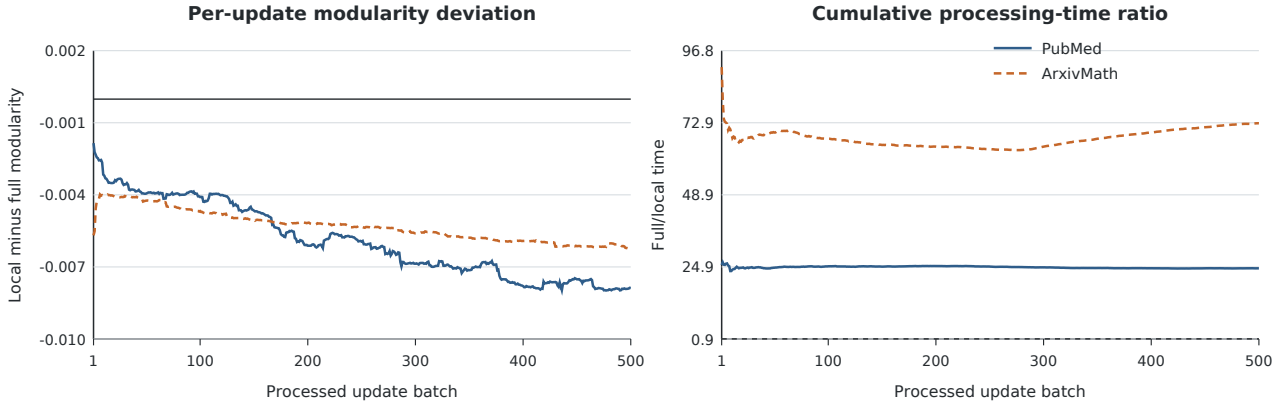


FIGURE 3. Leiden trajectories over 500 updates under 9:500. Left: per-batch modularity difference (ComNetX minus full Leiden), where negative values favor the full partition. Right: cumulative full/ComNetX processing-time ratio, where values above one favor ComNetX. Both curves cover the complete 500-update sequence.

TABLE 13. Fixed-window granularity control, 999:q. Here u is the actual number of nonempty updates, b the stored records per update, ΔQ final ComNetX-minus-baseline modularity, and S baseline/ComNetX time. DF-Leiden uses its native dynamic baseline. Each row is one execution per method. The repeated S^{2}CAG comparison at $q = 5, 10$ is given in Table 6.

Dataset	q	u	b	Leiden ΔQ	Leiden S	DF ΔQ	DF S
PubMed	5	5	8–9	-0.0043	5.24	+0.0206	0.32
PubMed	10	10	4–5	-0.0050	31.42	+0.0207	0.71
PubMed	20	20	2–3	-0.0060	39.21	+0.0277	0.61
PubMed	40	40	1–2	-0.0167	44.83	+0.0076	0.49
PubMed	80	44	1	-0.0179	45.57	+0.0077	0.51
PubMed	160	44	1	-0.0179	45.49	+0.0077	0.51
PubMed	320	44	1	-0.0179	45.43	+0.0077	0.51
ArxivMath	5	5	159–160	-0.0039	64.12	+0.0306	8.45
ArxivMath	10	10	79–80	-0.0046	99.58	+0.0309	10.05
ArxivMath	20	20	39–40	-0.0049	149.57	+0.0354	10.34
ArxivMath	40	40	19–20	-0.0065	232.46	+0.0215	10.71
ArxivMath	80	80	9–10	-0.0080	381.84	+0.0193	10.73
ArxivMath	160	160	4–5	-0.0084	556.49	+0.0207	10.42
ArxivMath	320	320	2–3	-0.0134	695.84	+0.0204	9.99

VIII. CONCLUSION†

ComNetX maintains a nested partition as edges change by restricting each detector call to a contracted graph. In the Leiden experiments the largest time savings occurred when small batches updated an established graph. Over 500 updates on PubMed and ArxivMath, the absolute modularity difference from full recomputation stayed below 0.008. The savings were smaller during graph growth. Compared with native DF-Leiden, ComNetX with the same backend was faster on ArxivMath and slower on PubMed, with higher final modularity and NMI on both. The repeated S^{2}CAG comparisons also favored ComNetX in measured time, but the full and local calls used different cluster-count rules and produced partitions at different granularities.

A practical next step is to test whether update-region and quotient sizes can guide the choice between local and full recomputation.

APPENDIX A BOUNDARY TERM FOR MODULARITY†

To relate modularity on the induced subgraph and on the full graph, assume $A_{ij} \geq 0$, $W = \sum_{i,j} A_{ij} > 0$, and $\gamma \geq 0$. Let $U = U_t^{\ell}$, $W_U = \sum_{i,j \in U} A_{ij} > 0$. Write $\boldsymbol{A}[U] = (A_{ij})_{i,j \in U}$ for the adjacency matrix of the induced subgraph on U, and let Z be a partition of U whose labels are disjoint from the

TABLE 14. Intermediate-window control, 99:50: 50 updates collectively covering approximately 1% of the stored record sequence. Values are baseline/ComNetX, with the same baseline and timing conventions as Table 12.

Dataset	Backend	Q	NMI	T (s)	Time ratio
PubMed	Leiden	0.7883/0.7809	0.2049/0.2052	24.831/1.028	24.16
PubMed	DF-Leiden	0.7401/0.7808	0.1875/0.2068	0.447/0.485	0.92
ArxivMath	Leiden	0.8855/0.8813	0.4181/0.4006	571.3/9.480	60.26
ArxivMath	DF-Leiden	0.8377/0.8810	0.3952/0.4002	13.637/1.506	9.05

TABLE 15. Graph growth under 0:10 from singleton initialization, including the first processed snapshot. Values are full Leiden/ComNetX; each row is one paired execution. The entire stored record sequence is divided into ten batches.

Dataset	Q	NMI	T (s)	Time ratio
Cora	0.8230/0.8150	0.4715/0.4642	0.985/0.640	1.54
ACM	0.8248/0.8232	0.2704/0.2687	0.934/0.801	1.17
CiteSeer	0.8932/0.8860	0.3302/0.3272	0.847/0.473	1.79
Patent	0.8567/0.8564	0.3617/0.3673	2.197/1.351	1.63
PubMed	0.7866/0.7797	0.2014/0.2094	5.652/3.972	1.42

fixed outside partition. Let C_Z combine Z with that retained outside partition. For each block a contained in U, let x_a and y_a be its outgoing and incoming degree within U, and define its boundary degrees by

$$p_a = \sum_{i\in a, j\notin U} A_{ij}, \qquad q_a = \sum_{i\notin U, j\in a} A_{ij}. \tag{24}$$

If $K_{V\setminus U}$ is the full-modularity contribution independent of Z, direct substitution in (5) gives

$$\begin{aligned} D(Z) &= Q_\gamma(\boldsymbol{A}, C_Z) - K_{V\setminus U} - \frac{W_U}{W} Q_\gamma(\boldsymbol{A}[U], Z) \\ &= \frac{\gamma}{W^2}\left[\frac{W - W_U}{W_U}\sum_a x_a y_a \right. \\ &\qquad \left. - \sum_a (x_a q_a + y_a p_a + p_a q_a)\right]. \end{aligned} \tag{25}$$

The internal-edge term is unchanged; the mismatch consists of the global-versus-local null-model normalization and the boundary degrees.

Write $\beta_{\text{out}} = \sum_a p_a$ and $\beta_{\text{in}} = \sum_a q_a$. Nonnegativity gives

$$-B_- \le D(Z) \le B_+, \tag{26}$$

where

$$\begin{aligned} B_+ &= \frac{\gamma}{W^2} W_U (W - W_U), \\ B_- &= \frac{\gamma}{W^2}\left[W_U(\beta_{\text{out}} + \beta_{\text{in}}) + \beta_{\text{out}}\beta_{\text{in}}\right]. \end{aligned} \tag{27}$$

For example, $\sum_a x_a y_a \le W_U^2$, $\sum_a x_a q_a \le W_U \beta_{\text{in}}$, and the analogous bounds for the remaining terms give (26).

For a fixed family $\mathcal{S}$ of partitions that keep the specified atoms intact, define $F(Z) = Q_\gamma(\boldsymbol{A}, C_Z) - K_{V\setminus U}$ and $L(Z) = (W_U/W) Q_\gamma(\boldsymbol{A}[U], Z)$. If Z_U maximizes L and Z_G maximizes F over $\mathcal{S}$, then

$$0 \le F(Z_G) - F(Z_U) \le B_- + B_+. \tag{28}$$

This bound compares exact optima only within that family; it does not bound loss against an unrestricted full-graph optimum or certify the output of a heuristic backend.

APPENDIX B MEMORY USE OF THE CURRENT IMPLEMENTATION[†]

The implementation stores Boolean masks for all fixed update regions and processes levels sequentially. Let $M_{\text{SpMM},\ell}$ and $M_{\mathcal{B},\ell}$ denote sparse-product and backend workspace at level ℓ. Let $M_{\text{feat},\ell}$ denote the additional storage for feature preparation and aggregation at that level, with $M_{\text{feat},\ell} = 0$ for topology-only calls. The implementation's total additional space is conservatively bounded by

$$\begin{aligned} M_{\text{extra}} = O\Big(Ln + \max\big\{s_t + n, \\ \max_\ell \big[e_\ell + h_\ell + q_\ell + M_{\text{feat},\ell} \\ + M_{\text{SpMM},\ell} + M_{\mathcal{B},\ell}\big]\big\}\Big). \end{aligned} \tag{29}$$

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